Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India

(Autonomous College Affiliated to University of Mumbai)

Mid Semester Examination

Oct. 2025

Course Name: Artificial Intelligence and Soft Computing — Course Code: **CS303/CE303**

Class: T. Y.B.Tech. — Max. Marks: 30

Branch: COMP/CSE — Semester: V

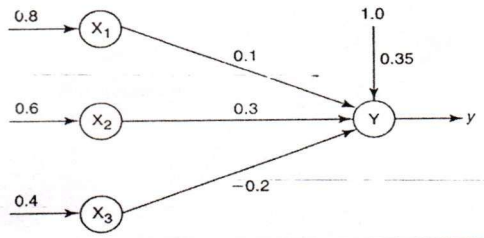
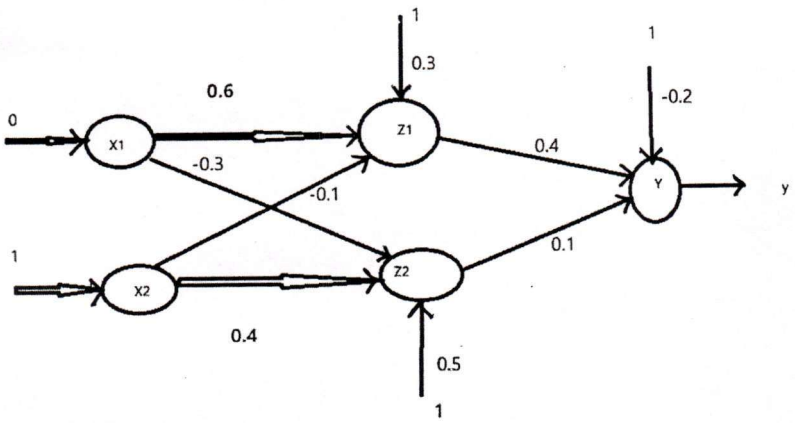
Time: 60 minutes

Instruction:

(1) Draw neat diagrams/designs.

(3) Assume suitable data if necessary

(4) Mention the question number clearly while writing the answer

QNo	Question	Marks	CO
Qu.1	a.) Distinguish between the following with suitable example 1. Simple Reflex Agent Vs. Goal based Agent 2. PEAS Vs PAGE b) Obtain the output of the neuron Y for the network shown in Figure using activation functions as binary sigmoidal.(take lambda=1) 	5 5	CO1
Qu.2	Using back-propagation network, find the new weights for the net shown in Figure. It is presented with the input pattern [0, 1] and the target output is 1. Use a learning rate alpha =0.25 and binary sigmoidal activation function with lambda=1 	10	CO2

Qu. 3	Explain the concept of Learning Vector Quantization (LVQ) and its role in classification problems. Consider an LVQ network with four input vectors belonging to two classes as shown below: Using the squared Euclidean distance measure, perform the LVQ weight update process with the following initial prototype weight vectors: Learning Rate =0.3 w1=[1 0 1 0] w2=[0 1 0 1] Show the final updated weight matrix after one compiete pass through the dataset.	10	CO2

Vector	Class
[1 0 1 0]	1
[0 1 0 1]	2
[1 1 0 0]	1
[0 0 1 1]	2
[0 1 1 1]	1